

Topic B15

Reproduction

Concept guide · Worked application · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

Syllabus map

B16.1	Asexual reproduction
B16.2	Sexual reproduction
B16.3	Reproduction in plants
B16.4	Human reproduction
B16.5	Sex hormones and birth control
B16.6	Sexually transmitted infections

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

1. Asexual and sexual reproduction

Asexual reproduction uses one parent, no gamete fusion and produces genetically identical offspring except for mutation. It is rapid and effective in stable conditions. Sexual reproduction fuses haploid gametes, produces variation and may improve adaptation when conditions change, but requires more time and energy.

Examiner insight

Compare using parent number, gametes, variation and environmental context.

2. Flowering plants

Anthers make pollen and ovaries contain ovules. Pollination transfers pollen from anther to stigma; fertilisation fuses male nucleus with egg nucleus in an ovule. A pollen tube grows through the style. After fertilisation, ovule becomes seed and ovary becomes fruit.

Examiner insight

Pollination is transfer; fertilisation is nuclear fusion.

3. Human reproduction

Testes make sperm and testosterone; ovaries make ova, oestrogen and progesterone. Fertilisation usually occurs in the oviduct; implantation occurs in the uterus. The placenta permits exchange without direct mixing of maternal and fetal blood; the umbilical cord connects fetus and placenta.

Examiner insight

Adaptations of sperm and egg should be linked to fertilisation and early development.

4. Cycles, contraception and STIs

FSH stimulates follicle development; oestrogen rebuilds the uterine lining and inhibits FSH; LH triggers ovulation; progesterone maintains the lining. Contraception acts through barriers, hormones, intrauterine devices or surgery. HIV is transmitted in body fluids and weakens immunity; condoms reduce transmission risk.

Examiner insight

Hormonal control uses interacting feedback, not a single hormone acting alone.

Worked application

Situation: An insect-pollinated flower population loses pollinators.

Prediction: Seed production decreases.

Reason: Fewer pollen grains reach compatible stigmas, reducing fertilisation.

Exam technique

High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.

Practice Paper

Topic B15

Reproduction

Instructions

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A - Multiple choice

Choose the best answer for each question.

1. Which statement best defines pollination? [1]
 - A transfer of pollen from anther to stigma
 - B fusion of nuclei of male and female gametes
 - C attachment of an early embryo to the uterine lining
 - D a haploid reproductive cell that fuses during fertilisation
2. Which statement best defines fertilisation? [1]
 - A attachment of an early embryo to the uterine lining
 - B fusion of nuclei of male and female gametes
 - C a haploid reproductive cell that fuses during fertilisation
 - D reproduction from one parent without gamete fusion
3. Which statement best defines implantation? [1]
 - A a haploid reproductive cell that fuses during fertilisation
 - B reproduction from one parent without gamete fusion
 - C attachment of an early embryo to the uterine lining
 - D transfer of pollen from anther to stigma
4. Which statement best defines gamete? [1]
 - A reproduction from one parent without gamete fusion
 - B transfer of pollen from anther to stigma
 - C fusion of nuclei of male and female gametes
 - D a haploid reproductive cell that fuses during fertilisation
5. Which statement best defines asexual reproduction? [1]
 - A reproduction from one parent without gamete fusion
 - B transfer of pollen from anther to stigma
 - C fusion of nuclei of male and female gametes
 - D attachment of an early embryo to the uterine lining
6. Which statement about asexual and sexual reproduction is correct? [1]
 - A Fusion of male and female nuclei.
 - B Asexual reproduction uses one parent, no gamete fusion and produces genetically identical offspring except for mutation.
 - C Implantation of an embryo.
 - D Release of an ovum.
7. Which statement about flowering plants is correct? [1]
 - A Implantation of an embryo.
 - B Release of an ovum.
 - C Anthers make pollen and ovaries contain ovules.
 - D A diploid body cell.

8. Which statement about human reproduction is correct? [1]
- A Release of an ovum.
 - B A diploid body cell.
 - C Reproduction that always creates variation.
 - D Testes make sperm and testosterone; ovaries make ova, oestrogen and progesterone.
9. Which statement about cycles, contraception and stis is correct? [1]
- A FSH stimulates follicle development; oestrogen rebuilds the uterine lining and inhibits FSH; LH triggers ovulation; progesterone maintains the lining.
 - B A diploid body cell.
 - C Reproduction that always creates variation.
 - D Fusion of male and female nuclei.
10. Which prediction is most appropriate when an insect-pollinated flower population loses pollinators? [1]
- A Fusion of male and female nuclei.
 - B Release of an ovum.
 - C Seed production decreases.
 - D Reproduction that always creates variation.

Section B · Structured questions

11 Biologists use precise vocabulary when studying reproduction.

(a) Define pollination.

[2]

(b) Define fertilisation.

[2]

(c) Distinguish implantation from the incorrect idea: “release of an ovum”.

[2]

(d) State two important features of asexual and sexual reproduction.

[2]

[Total: 8]

12 An insect-pollinated flower population loses pollinators.

(a) State the most likely biological outcome.

[2]

(b) Explain the outcome using a linked biological mechanism.

[3]

(c) Predict what would happen if the relevant change were reversed. [2]

(d) State one assumption made in this prediction. [1]

[Total: 8]

13 A student records relative hormone level under three conditions.

condition	relative hormone level
day 7	3
day 14	18
day 21	4

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

(c) Explain the pattern using biological knowledge. [3]

(d) State one limitation of drawing a conclusion from these data. [1]

[Total: 8]

14 A student plans to investigate seed germination at different temperatures.

(a) Identify the independent variable. [1]

(b) Identify the dependent variable. [1]

(c) State two control variables. [2]

(d) Describe how to obtain reliable results across a suitable range. [3]

(e) Suggest one specific improvement. [1]

[Total: 8]

15 Apply your knowledge of reproduction to an unfamiliar context.

(a) Define gamete. [2]

(b) Explain the biological importance of human reproduction. [3]

(c) Correct this statement: “reproduction that always creates variation”. [2]

(d) State one further question that could be investigated. [1]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

Solutions · Multiple choice

- 1 A. Transfer of pollen from anther to stigma.
- 2 B. Fusion of nuclei of male and female gametes.
- 3 C. Attachment of an early embryo to the uterine lining.
- 4 D. A haploid reproductive cell that fuses during fertilisation.
- 5 A. Reproduction from one parent without gamete fusion.
- 6 B. Compare using parent number, gametes, variation and environmental context.
- 7 C. Pollination is transfer; fertilisation is nuclear fusion.
- 8 D. Adaptations of sperm and egg should be linked to fertilisation and early development.
- 9 A. Hormonal control uses interacting feedback, not a single hormone acting alone.
- 10 C. Fewer pollen grains reach compatible stigmas, reducing fertilisation.

Solutions · Structured questions

11. (a) transfer of pollen from anther to stigma [2]. (b) fusion of nuclei of male and female gametes [2]. (c) attachment of an early embryo to the uterine lining; not release of an ovum [2]. (d) Credit any two accurate features from the guide: Asexual reproduction uses one parent, no gamete fusion and produces genetically identical offspring except for mutation. It is rapid and effective in stable conditions. Sexual reproduction fuses haploid gametes, produces variation and may improve adaptation when conditions change, but requires more time and energy. [2].
12. (a) Seed production decreases [2]. (b) Fewer pollen grains reach compatible stigmas, reducing fertilisation [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].
13. (a) $18 - 3 = 15$ [2]. (b) The values change from 3 to 4, with the highest value 18. [2]. (c) The lh surge around day 14 triggers ovulation [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].
14. (a) Temperature [1]. (b) Percentage germination [1]. (c) Same species, seed number, water, oxygen and time [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Use replicates and thermostatically controlled incubators [1].
15. (a) a haploid reproductive cell that fuses during fertilisation [2]. (b) Award linked points from: Testes make sperm and testosterone; ovaries make ova, oestrogen and progesterone. Fertilisation usually occurs in the oviduct; implantation occurs in the uterus. The placenta permits exchange without direct mixing of maternal and fetal blood; the umbilical cord connects fetus and placenta. [3]. (c) Correct statement: asexual reproduction means reproduction from one parent without gamete fusion [2]. (d) Any testable relevant biological question [1].

Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.